

# LandWeb Study-Area Vegetation

SCANFI species composition over the LandWeb study area

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## Warning

### Draft report: pending human review

Portions of this document were drafted with the help of generative AI (large language models) and have not yet been fully reviewed by the co-authors. Content may be incomplete, inaccurate, or misattributed. This notice will be removed once a complete review has been carried out.

## 1 Overview

This report summarizes the SCANFI v2 (2020) (Guindon et al. 2026) per-species **crown-closure prevalence** over the **full LandWeb study area** (3,016,183 km<sup>2</sup>, 3226467208 forested 30 m cells). For each SCANFI source species that maps to a simulated LandWeb species it reports the number (and %) of forested study-area cells carrying *any* cover.

Its primary purpose is to document the evidence behind the **THUJ\_PLI/TSUG\_HET** → **Abie\_spp merge** in **LandWeb\_preamble**: western redcedar and western hemlock are a *localized* SCANFI over-attribution in the SW Alberta Rockies, not a study-area-wide signal.

## 2 Study area composition

Table 1: SCANFI species prevalence over the LandWeb study area, by cells with any cover (sorted). **LandWeb species** is the simulated species each SCANFI code merges into.

| SCANFI code  | Common name              | LandWeb species | Cells (cover > 0) | % of study area |
|--------------|--------------------------|-----------------|-------------------|-----------------|
| PICE_MAR     | black spruce             | Pice_mar        | 1,548,758,984     | 48.002          |
| LARI_LAR     | tamarack                 | Lari_spp        | 924,800,678       | 28.663          |
| POPU_TRE     | trembling aspen          | Popu_spp        | 897,899,898       | 27.829          |
| PINU_BAN     | jack pine                | Pinu_spp        | 840,128,756       | 26.039          |
| PICE_GLA     | white spruce             | Pice_gla        | 701,495,401       | 21.742          |
| BETU_PAP     | white birch              | Popu_spp        | 616,449,196       | 19.106          |
| POPU_BAL     | balsam poplar            | Popu_spp        | 290,988,009       | 9.019           |
| PINU_CON_LAT | lodgepole pine           | Pinu_spp        | 242,557,446       | 7.518           |
| ABIE_BAL     | balsam fir               | Abie_spp        | 152,552,888       | 4.728           |
| ABIE_LAS     | subalpine fir            | Abie_spp        | 89,885,903        | 2.786           |
| PICE_ENG_GLA | Engelmann x white spruce | Pice_gla        | 54,901,702        | 1.702           |
| PICE_ENG     | Engelmann spruce         | Pice_gla        | 44,336,103        | 1.374           |
| PSEU_MEN     | Douglas-fir              | Pseu_men        | 10,487,529        | 0.325           |
| LARI_OCC     | western larch            | Lari_spp        | 5,868,751         | 0.182           |
| PSEU_MEN_GLA | Douglas-fir (interior)   | Pseu_men        | 5,430,224         | 0.168           |
| THUJ_PLI     | western redcedar         | Abie_spp        | 5,248,943         | 0.163           |
| TSUG_HET     | western hemlock          | Abie_spp        | 4,647,291         | 0.144           |
| PINU_CON_CON | shore pine               | Pinu_spp        | 3,173             | 0.000           |

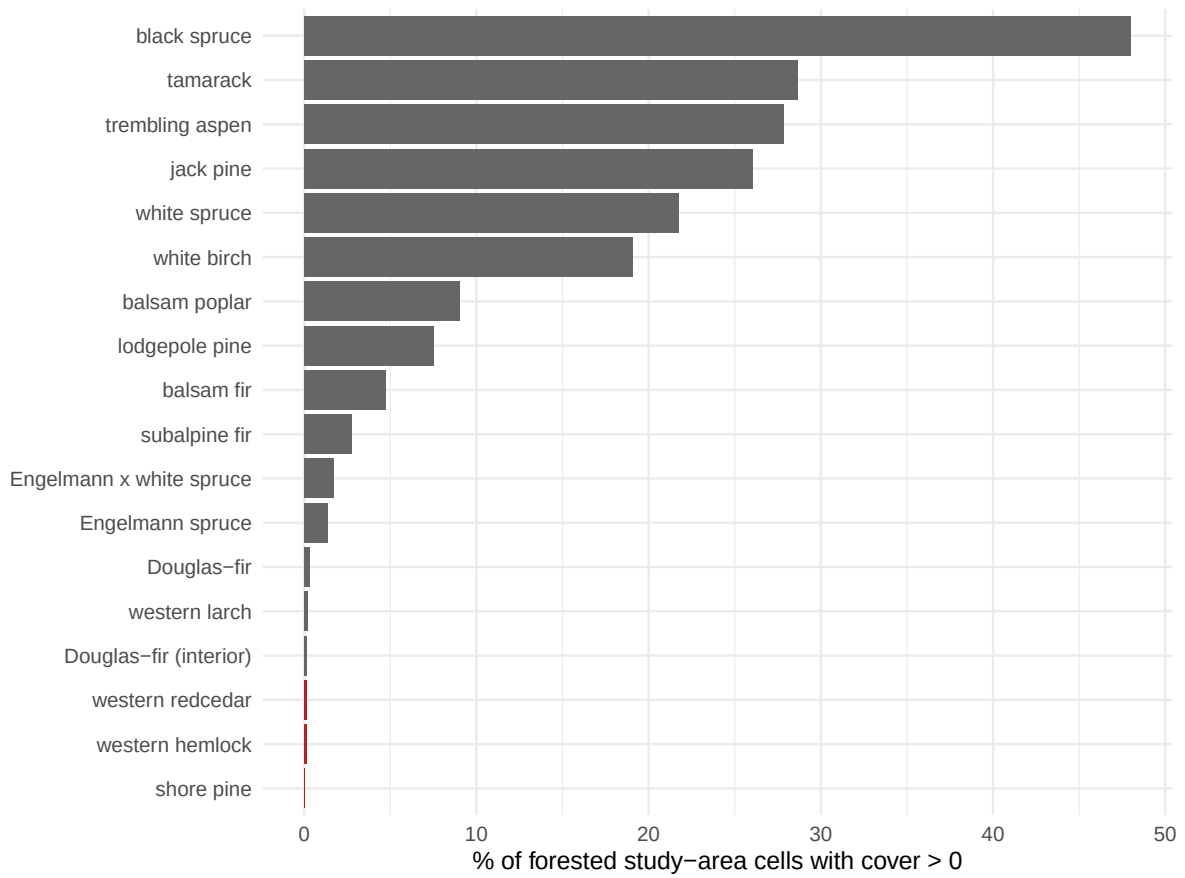


Figure 1: Prevalence (% of forested study-area cells with cover > 0) by SCANFI species; coastal-attribution suspects in red.

The composition is a textbook western-boreal signature — black spruce dominant, then tamarack, aspen, jack pine, and white spruce.

### 3 Coastal over-attribution (the merge rationale)

Table 2: Coastal-attribution suspects: study-area prevalence and substantial-cover ( $\geq 10\%$ ) extent.

| SCANFI code  | Common name      | % study area (cover $> 0$ ) | % study area ( $\geq 10\%$ ) | Max cover % |
|--------------|------------------|-----------------------------|------------------------------|-------------|
| THUJ_PLI     | western redcedar | 0.163                       | 0.0373                       | 61          |
| TSUG_HET     | western hemlock  | 0.144                       | 0.0410                       | 70          |
| PINU_CON_CON | shore pine       | 0.000                       | 0.0000                       | 24          |

Western hemlock (TSUG\_HET) covers only **0.14%** of the LandWeb study area, and western redcedar (THUJ\_PLI) **0.16%** — rounding errors at the study-area scale. By contrast, a stand-alone count over the **Spray Lake FMA** (SW Alberta Rockies) put western hemlock on **~25%** of pixels: a  $\sim 175\times$  concentration in the SW corner, exactly where SCANFI bleeds coastal species across the BC boundary. Where these species do appear they can reach high cover (up to 70% hemlock), but the footprint is tiny and localized.

Because both fold into **Abie\_spp** (the closest shade-tolerant softwood), the merge:

- removes a stand-alone species that has no usable known-age / non-zero-biomass cohorts in the SW — the cause of the **Biomass\_borealDataPrep** non-conformable (“code 235”) crash; and
- relabels only  $\sim 0.3\%$  of the study area’s cover, leaving the rest untouched.

PINU\_CON\_CON (shore pine, the coastal *Pinus contorta* variety) is effectively absent across the study area (3,173 cells), so it is left merged into **Pinu\_spp**, where — like all pines — it is harmless.

## References

- Guindon, L., D. L. P. Correia, F. Manka, and B. Smiley. 2026. *SCANFI V2: Spatialized Canadian National Forest Inventory Data Product V2*. Natural Resources Canada, Canadian Forest Service, Laurentian Forestry Centre, Quebec, Canada. <https://doi.org/10.23687/07653869-f303-46c2-a04e-9ab479b73cbf>.
- Guindon, Luc, Francis Manka, David L. P. Correia, et al. 2024. “A New Approach for Spatializing the Canadian National Forest Inventory (SCANFI) Using Landsat Dense Time Series.” *Canadian Journal of Forest Research* 54 (7): 793–815. <https://doi.org/10.1139/cjfr-2023-0118>.

## Provenance

- **Source:** SCANFI v2 species crown-closure layers (Guindon et al. 2026), derived via the SCANFI approach (Guindon et al. 2024) (`SCANFI_spsCC_<code>_2020_v2_20260119.tif`), masked to the LandWeb study area (the LTHFC v10 union, `landweb_lthfc_v10`).
- **Data file:** `outputs/extended_analyses/scanfi_lthfc_v10_2020_veg_summary.csv`
- **Method:** `R/scanfi_veg_summary.R`, built by the gated `landweb.extended_analyses` target. The scan is multi-hour over the  $\sim 3.0$  M km<sup>2</sup> study area, so the result is cached as a `format = "file"` target and reused unless the CSV is deleted or the SCANFI/LTHFC version stamp changes.